

WORKING PAPER IN LABOUR ECONOMICS

CEDS Paris - Version 2, August 2026

Labor Cost Lab: An Open-Source Comparative Observatory of Labour Costs, Contribution Reliefs and Socio-Fiscal Marginal Returns

*Six national modules and a purchasing-power-parity
comparison layer, from the French Labour Cost Lab*

Hugo Spring-Ragain

Economist, CEDS Paris

hugosragain.github.io/labor-cost-lab

JEL Classification: J30, J32, J38, H24, H55, C80

Keywords: labour costs, social contributions, payroll taxes, contribution reliefs, RGDU,

tax wedge, income tax, in-work benefits, socio-fiscal marginal return, purchasing power parity, cross-country comparison, simulation, open-source, France, Germany, Belgium, Switzerland, Netherlands, United Kingdom

This working paper documents an open-source research tool. It supersedes the *French Labour Cost Lab* working paper (V1, May 2026), which is retained as the methodological reference for the France module. It does not represent the official position of any institution. All errors remain the author's own.

Abstract

This working paper presents the *Labor Cost Lab*, an open-source comparative observatory that reconstructs how socio-fiscal rules transform an employer cost into a net wage. The project originates in the *French Labour Cost Lab* (FLCL), a single-country dashboard documented in a first working paper, and generalizes its architecture into a multi-country platform. The Lab now comprises six national modules – France, Germany, Belgium, Switzerland, the Netherlands and the United Kingdom – and a transversal comparison module that places them on a common wage grid expressed in international dollars at purchasing power parity.

Each national module follows the same logic: a reference wage grid, an explicit and versioned set of regulatory parameters, a calculation engine, and a harmonized set of outputs covering gross wage, net wage, total employer cost, employee and employer contributions, statutory reliefs, effective rates, socio-fiscal wedges and the marginal return to a gross wage increase. The France module retains the deepest institutional resolution: it queries the public Mon-entreprise/URSSAF engine over a grid running from 0.8 to 6.0 times the statutory minimum wage with a step of 0.01 SMIC, across a combinatorial profile system, and reports the Reduction Generale Degressive Unique (RGDU), a reference socio-fiscal module including estimated personal income tax and estimated prime d’activite, and a family of labour cost efficiency indicators (the FLCL Index).

The comparison module does not redefine any national parameter. It converts each harmonized wage point into local currency using the World Bank private-consumption purchasing power parity factor, then calls each national engine directly, without interpolation, for an explicitly documented reference profile per country. Wage points falling below a country’s statutory minimum are flagged as hypothetical rather than silently dropped.

The contribution of the Lab is fourfold: reproducibility, through documented pipelines, versioned regulatory parameters and open datasets; wage-scale granularity, which exposes non-linearities and threshold effects invisible in aggregate indicators or isolated case-type calculations; visibility of marginal incentives, on both the employer and the employee side; and comparability, through a harmonization protocol that separates what is a genuine institutional difference from what is a currency or price-level artefact. The Lab documents accounting and institutional incidence. It does not estimate economic incidence, behavioural responses or employment effects, and it is not an official payroll calculator.

Keywords: labour costs, social contributions, payroll taxes, contribution reliefs, RGDU, tax wedge, income tax, in-work benefits, socio-fiscal marginal return, purchasing power parity, cross-country comparison, simulation, open-source.

JEL Classification: J30, J32, J38, H24, H55, C80.

1. Introduction

The structure of labour costs is a subject of considerable economic interest, both for research and for policy. Continental European systems in particular combine statutory employer and employee contribution schedules with targeted reliefs, income taxation and in-work benefits, producing a labour cost schedule that is highly non-linear along the wage distribution. France is an extreme case: an extensive set of degressive employer contribution reliefs concentrated near the statutory minimum wage interacts with executive status, territorial regime, occupational injury classification, firm size and working time.

Despite this complexity, most public discussion of labour costs relies on one of two types of evidence: aggregate macroeconomic indicators, which mask within-distribution heterogeneity, or isolated case-type calculations, which are rarely reproducible and typically fail to capture marginal effects along the wage scale. Academic contributions have long documented the importance of considering the full schedule of effective contribution rates when analysing the employment effects of payroll taxes and the incidence of reliefs (5; 7; 3). In the French policy debate, the question of where along the wage scale reliefs phase out, and at what marginal cost, has become central (2). Yet practitioners, students and policy analysts still lack access to a transparent, granular and reproducible tool for visualizing these mechanisms – and, when they want to compare countries, they face an additional obstacle: nominal wages are not comparable across price levels, and institutional categories do not map onto one another.

The *Labor Cost Lab* addresses both gaps. It began as the *French Labour Cost Lab* (FLCL), a single-country dashboard simulating gross wages, net wages, employer costs, contributions and reliefs across a fine-grained French wage grid. That tool is documented in a first working paper (12) and is now the France module of a larger platform. The Lab currently hosts six national modules – France, Germany, Belgium, Switzerland, the Netherlands and the United Kingdom – and a transversal comparison module that evaluates all six on a wage grid harmonized in international dollars at purchasing power parity.

The change of scope is not cosmetic. A single-country dashboard answers the question: *how does this system transform an employer cost into a net wage along the wage scale?* A comparative observatory answers a different one: *which features of that transformation are specific to one institutional architecture, and which are common to systems of the same family?* Answering the second question requires a harmonization protocol that is itself an object of methodological choice, and the Lab therefore treats the comparison layer as documented infrastructure rather than

as an incidental output.

The contribution of the *Labor Cost Lab* is fourfold. First, **reproducibility**: simulations are generated from documented pipelines with explicit, versioned regulatory parameters, separating calculated data, regulatory assumptions and documentary sources. Second, **wage-scale granularity**: the France module covers 0.8 to 6.0 SMIC with a step of 0.01 SMIC, enabling the detection of non-linearities and discontinuities that case-type calculations cannot reveal. Third, **visibility of marginal incentives**: the Lab shows how contribution reliefs, employee contributions, income taxation and in-work benefits jointly shape the marginal return to gross wage increases. Fourth, **comparability**: the comparison module harmonizes wage levels by purchasing power parity and calls each national engine directly, so that cross-country differences reflect institutional rules rather than currency conversion or price-level effects.

The remainder of the paper is organized as follows. Section 2 reviews the conceptual foundations of labour cost analysis and the distinction between accounting and economic incidence. Section 3 describes the architecture of the Lab and the transition from the FLCL to a multi-country platform. Section 4 documents the France module. Section 5 documents the five other national modules. Section 6 documents the comparison module and its harmonization protocol. Section 7 defines the indicators, including the FLCL Index. Section 8 discusses descriptive results, including the 1 June 2026 minimum wage revalorisation. Section 9 describes the dashboard architecture. Section 10 discusses intended uses. Section 11 enumerates methodological limitations. Section 12 concludes.

2. Conceptual Background

2.1. Decomposition of the labour cost

In the French institutional framework, the total cost of labour borne by the employer decomposes as follows:

$$C_{\text{emp}} = w_{\text{brut}} + \text{Cot}_{\text{emp}}^{\text{gross}} - \text{RGDU} \quad (1)$$

where C_{emp} denotes the total employer cost, w_{brut} the gross wage, $\text{Cot}_{\text{emp}}^{\text{gross}}$ the statutory employer social contributions before relief, and RGDU the contribution relief received by the employer under the degressive unified relief scheme. Defining net employer contributions as

$$\text{Cot}_{\text{emp}}^{\text{net}} = \text{Cot}_{\text{emp}}^{\text{gross}} - \text{RGDU}, \quad (2)$$

equation (1) is equivalently:

$$C_{\text{emp}} = w_{\text{brut}} + \text{Cot}_{\text{emp}}^{\text{net}}. \quad (3)$$

The net wage received by the employee is:

$$w_{\text{net}} = w_{\text{brut}} - \text{Cot}_{\text{sal}}, \quad (4)$$

where Cot_{sal} represents employee social contributions deducted from the gross wage.

This decomposition generalizes to the other national modules with one substitution: the relief term is replaced by the country's own statutory reduction mechanism where one exists (the Belgian structural reduction, for instance), and set to zero where none does. The Dutch case requires an additional convention, discussed in Section 5.4, because employee national insurance contributions are bundled into the first bracket of the income tax schedule rather than withheld separately.

These accounting identities define the *institutional incidence* of social contributions: they describe who formally pays what, as determined by the legal and regulatory framework. This is the primary object of analysis in the Lab.

2.2. Socio-fiscal wedge

The socio-fiscal wedge before tax, or *coin social*, is the total contribution burden extracted from the employment relationship, expressed as the gap between employer cost and employee net wage before income tax:

$$W = C_{\text{emp}} - w_{\text{net}} = \text{Cot}_{\text{emp}}^{\text{net}} + \text{Cot}_{\text{sal}}. \quad (5)$$

The total wedge after tax adds the income tax actually due on the wage for the reference profile:

$$W^{\text{tot}} = C_{\text{emp}} - (w_{\text{net}} - \text{IR}). \quad (6)$$

The distinction matters for comparison. Systems differ in how much of the levy is routed through social contributions rather than income taxation, so a comparison

restricted to the pre-tax wedge would systematically flatter countries that tax labour mainly through the income tax schedule. Reporting both wedges is the minimum condition for a defensible cross-country statement.

2.3. Economic incidence versus accounting incidence

A fundamental principle of public finance is that the formal legal incidence of a tax does not necessarily coincide with its economic incidence, that is, who ultimately bears the burden in terms of purchasing power (6; 5). In labour markets, economic incidence depends on the relative elasticities of labour supply and labour demand, the role of minimum wage regulations, collective bargaining institutions and market structure (8; 11).

In France, the SMIC creates a wage floor that constrains downward adjustment of gross wages, which may affect the distribution of the incidence burden at the bottom of the wage distribution (7). Degressive contribution reliefs targeted at low wages further imply that the effective labour cost schedule is not proportional to gross wages, generating complex employment incentive effects (3; 9).

The *Labor Cost Lab* does not attempt to estimate economic incidence. It simulates accounting incidence and institutional schedules. It provides an open computational infrastructure that can support further empirical work, but it does not estimate behavioural responses or causal employment effects.

2.4. Contribution reliefs and marginal cost

France has implemented employer contribution relief systems since the mid-1990s, progressively consolidated into the current Reduction Generale Degressive Unique (RGDU). The RGDU is maximal at wages equal to the SMIC and decreases as the ratio of gross wage to SMIC increases, reaching zero at a regulatory threshold. This degressive profile implies that the effective employer contribution rate is substantially lower than the statutory rate at low wages and converges toward the statutory rate at higher wages.

The RGDU introduces a non-linearity in the effective labour cost schedule. Within the phase-out zone, the marginal employer cost of a wage increase can exceed the average effective cost, because a wage increase simultaneously raises statutory contributions and reduces the relief received. This mechanism is central to the France module, and its analytical zone extends considerably further up the wage distribution than the thresholds usually invoked in public debate – which is precisely why a grid restricted to the neighbourhood of the minimum wage is insufficient.

2.5. From labour cost to socio-fiscal return

For employees, the return to an increase in gross wage is not determined by employee social contributions alone. Disposable income is also affected by personal income tax and by the withdrawal of income-tested transfers. In France, the prime d'activité is particularly relevant around low and middle earnings, because it directly supplements labour income and is progressively reduced as resources increase.

This creates a second layer of marginal effects. The employer faces a marginal labour cost schedule shaped by employer contributions and reliefs. The employee faces a marginal disposable income schedule shaped by employee contributions, income taxation and benefit withdrawal. The Lab therefore reports a *socio-fiscal return* measure, designed to show how much of an additional euro of gross wage is transformed into additional disposable income for a reference profile.

2.6. Comparability across institutional architectures

Cross-country comparison of labour costs raises three distinct problems, which the Lab treats separately rather than collapsing into a single conversion step.

The first is the **price-level problem**. A given nominal wage does not correspond to the same standard of living in Zurich and in Lisbon. Comparing systems at equal nominal wages therefore evaluates them at structurally different positions in their own wage distributions. The Lab addresses this by harmonizing on purchasing power parity, following the logic used in the OECD's *Taxing Wages* framework (10).

The second is the **boundary problem**. What counts as a social contribution rather than a tax, or as a mandatory levy rather than a private arrangement, is a national legal convention. Occupational pension contributions are statutory in some systems and negotiated in others; national insurance may be withheld separately or merged into the income tax schedule. Rather than imposing a single normative boundary, the Lab documents the boundary retained for each module and flags it explicitly wherever it departs from the modal case.

The third is the **reference profile problem**. Every effective rate is conditional on a household and employment configuration. The Lab therefore fixes a single, explicitly stated reference profile per country – broadly, a single, childless, full-time employee – and treats it as a documented convention, not as a representative agent.

3. Architecture of the Lab

3.1. From the French Labour Cost Lab to the Labor Cost Lab

The *French Labour Cost Lab* was designed as a single-country dashboard: one wage grid, one calculation engine, one institutional system. Its architecture, however, separated three layers that turned out to be generalizable – a wage grid, a parameter set and an indicator layer. The transition to the *Labor Cost Lab* consists in making that separation explicit and instantiating it once per country.

Three consequences follow. First, the FLCL becomes the France module rather than the project itself; its methodology, its RGDU treatment and its socio-fiscal module are unchanged in substance and continue to be documented in the original working paper (12). Second, the indicator layer becomes a shared contract: every module must produce the same core outputs, so that a comparison is possible without post-hoc reconciliation. Third, the comparison itself becomes a module, with its own parameters, its own sources file and its own limitations section, rather than an informal juxtaposition of national results.

3.2. Common structure of a national module

Each national module is built from four components.

A reference wage grid. Wages are expressed relative to a national anchor – a multiple of the statutory minimum wage where one exists, or a reference wage otherwise – and evaluated at a fixed step.

A versioned parameter set. Contribution rates, thresholds, relief formulas, tax schedules and calculation conventions are stored as explicit parameters with a version identifier and an effective date, so that any published figure can be traced to the regulatory state that produced it.

A calculation engine. For France, the engine is the public Mon-entreprise/URSSAF API (14). For the other modules, the engine is a documented implementation of the statutory schedules from official sources.

A harmonized output grid. Every module returns the same core series: gross wage, employee contributions, net wage before tax, income tax, net wage after tax, employer contributions, total employer cost, effective rates, socio-fiscal wedges and marginal indicators.

3.3. Data, parameters and sources

The project deliberately separates three categories of artefact: computed datasets (CSV wage grids, one row per wage point and profile), regulatory parameters (JSON files carrying rates, thresholds and calculation conventions per country and per year), and documentary sources (JSON files listing the legal texts, official databases and administrative documents used to construct each module). This separation is what allows a reader to distinguish an assumption from a legal provision from a simulation output – the minimal condition for a tool of this kind to be auditable.

All artefacts are downloadable from the interface, and the code used to build the datasets and the site is open.

4. The France Module

4.1. The Mon-entreprise/URSSAF calculation engine

The French simulations are generated by querying the public API of *Mon-entreprise*, the official URSSAF simulator (<https://mon-entreprise.urssaf.fr>). This API exposes a calculation engine implementing current French social contribution rules, including statutory contribution rates, exemption thresholds, degressive relief formulas and territorial specificities.

The choice to rely on this external engine, rather than reproducing contribution rules manually, reflects a methodological decision. Manual reproduction of the French social contribution system is error-prone given the complexity of the regulatory framework. By delegating the core payroll contribution calculation to the public engine, the project concentrates its own contribution on wage grid design, profile combinations, indicator construction and visualization.

4.2. Wage grid

Simulations are run over a fine-grained grid of gross monthly wages expressed as multiples of the SMIC:

- Minimum wage point: **0.8 SMIC**;
- Maximum wage point: **6.0 SMIC**;
- Step: **0.01 SMIC**;
- Number of wage points: **521**.

The extension of the upper bound from 3.5 to 6.0 SMIC, relative to the first version of the tool, is substantive rather than cosmetic. It ensures that the entire phase-out zone of the RGDU is contained strictly inside the grid, that the effective employer

contribution rate can be observed converging toward its statutory asymptote, and that executive-status contribution features located above the social security ceiling are visible.

The SMIC reference used to convert multiples into euro amounts is a versioned parameter. Three values are relevant for the 2026 regulatory year: 1,801.80 euros per month until 31 December 2025, 1,823.03 euros from 1 January 2026, and 1,867.02 euros from 1 June 2026 following the automatic revalorisation of 2.41 percent triggered by the inflation clause of article L. 3231-5 of the Labour Code and fixed by the arrete of 22 May 2026 ([Arrete du 22 mai 2026](#)). The parameter file declares which value anchors a given dataset version; published figures should always be read together with that declaration.

4.3. Combinatorial profile system

The interface exposes five profile dimensions.

1. **Employee status.** Non-executive (non-cadre) or executive (cadre).
2. **Territorial regime.** Standard (outside Alsace-Moselle) or Alsace-Moselle.
3. **Occupational injury risk (AT/MP).** Standard AT/MP, AT/MP 1 percent, AT/MP 4 percent, or a support-functions scenario.
4. **Firm size.** Fewer than 50 employees, or 50 employees and above.
5. **Working time.** Full-time, 80 percent part-time, or 50 percent part-time.

The downloadable core dataset is indexed on the three original dimensions – status, territorial regime and AT/MP scenario – yielding 16 profiles ($2 \times 2 \times 4$) evaluated over the 521-point grid, that is 8,336 observations. Firm size and working time are applied as additional parameters at query time in the interface. The AT/MP dimension should be read as a risk-rate scenario, not as a precise occupation or sector; the support-functions scenario in particular relies on a rule flagged as experimental by the API and should be interpreted with caution.

4.4. Reference socio-fiscal module

The socio-fiscal module is not a general microsimulation model. It is a reference calculation designed to illustrate how personal income tax and the prime d'activite affect the marginal disposable return to wage increases. The default reference profile is:

- single individual;
- one tax unit (une part de quotient familial);
- no children;
- no other income;
- no housing benefit;
- no tax credits or reductions;
- standard 10 percent professional-expense allowance, bounded by a floor of 509 euros and a ceiling of 14,555 euros;
- 2026 personal income tax schedule applied to 2025 income, with marginal brackets of 0, 11, 30, 41 and 45 percent;
- decote and low-tax recovery threshold included;
- prime d'activite estimated using a simplified formula for a single person.

The calculation is indicative and should not be interpreted as an official tax or CAF assessment. Two points deserve emphasis. First, the amount computed here is the tax definitively due for this profile, not the monthly withholding actually applied on a payslip: the prelevement a la source operates through a personalized rate transmitted by the DGFIP or, failing that, the default rate of article 204 H of the General Tax Code, both of which are advances reconciled the following year. Second, the prime d'activite was reformed in 2026, which affects the withdrawal profile around low earnings and therefore the shape of the socio-fiscal return curve in that range.

4.5. Internal consistency checks

To verify that the simulated dataset is internally coherent, the module reports a set of consistency checks across all observations. These checks are not an external validation against payroll records. They verify that the main accounting identities used to construct the indicators hold within the simulated dataset.

Table 1: Internal consistency checks

Check	Accounting identity
Net wage identity	$w_{\text{net}} = w_{\text{brut}} - \text{Cot}_{\text{sal}}$
Employer cost identity	$C_{\text{emp}} = w_{\text{brut}} + \text{Cot}_{\text{emp}}^{\text{net}}$
Socio-fiscal wedge identity	$C_{\text{emp}} - w_{\text{net}} = \text{Cot}_{\text{emp}}^{\text{net}} + \text{Cot}_{\text{sal}}$
Cost-to-net ratio identity	$\rho = C_{\text{emp}}/w_{\text{net}}$
RGDU rate identity	$\tau_{\text{RGDU}} = \text{RGDU}/w_{\text{brut}}$

These checks are computed over the full set of simulated observations, and the

maximum and mean absolute deviations associated with each identity are reported. This provides a direct diagnostic of whether the derived indicators are coherent with the underlying simulation outputs before any economic interpretation is proposed.

5. The Other National Modules

The five other modules follow the common structure of Section 3.2. They are at different stages of maturity: France is the reference implementation, Belgium is a full 2026 socio-fiscal module, and Germany, Switzerland, the Netherlands and the United Kingdom are documented prototypes. Each is presented below with its reference profile and its principal modelling conventions.

5.1. Germany

The German module implements employee and employer social contributions and employer cost, with the income tax layer treated separately. The reference profile is an employee affiliated to public health insurance, childless, and located outside Saxony – two conditions that matter directly, since the childless supplement to long-term care insurance and the Saxon split of that contribution both alter the employee-employer allocation at unchanged gross wage.

5.2. Belgium

The Belgian module implements the 2026 socio-fiscal framework: ONSS employee and employer contributions, the employer structural reduction, and the precompte professionnel. The reference profile is a private-sector employee, which is currently the only profile modelled. Belgium is analytically instructive alongside France because it combines a comparatively high statutory contribution level with its own low-wage employer reduction, so the two systems can be read as different parameterizations of the same policy instrument.

5.3. Switzerland

The Swiss module implements AVS/AI/APG, unemployment insurance, a standardized occupational pension (LPP) assumption, accident insurance, and cantonal preparation for the 26 cantons. Two features require explicit convention. First, Switzerland has no single national tax scale: withholding tax is set at cantonal level, so the reference value reported is a population-weighted average across the 26 cantons, using 2024 permanent resident population from the Federal Statistical Office (4). Second, the tariff retained is A0N – single, no children, no church tax

– and the LPP and accident-insurance assumptions are standardized rather than employer-specific.

5.4. The Netherlands

The Dutch module implements the loonheffing and employer contributions. Its distinguishing feature is a boundary convention that must be stated explicitly: the Netherlands does not withhold a separate employee social contribution, because the AOW (old age), Anw (survivors) and Wlz (long-term care) national insurance contributions are bundled into the rate of the first Box 1 bracket. In this module the Dutch net wage before tax therefore equals the gross wage, and the employee contributions indicator equals zero – not because no social contribution is due, but because it is accounted for within the loonheffing. Any cross-country reading of the pre-tax wedge must take this convention into account.

The loonheffing is computed for a single, childless employee via the 2026 Box 1 schedule (35.75 percent up to 38,883 euros; 37.56 percent up to 78,426 euros; 49.50 percent above), reduced by two tax credits each computed on its own scale: the algemene heffingskorting and the arbeidskorting. The reference profile assumes a permanent contract (low WW rate) and a large employer (high Aof rate). The Whk, which is employer- and sector-specific, is not modelled, nor are occupational pension contributions, which are not set by law, nor partner-dependent tax credits.

5.5. United Kingdom

The United Kingdom module implements income tax (PAYE), Class 1 National Insurance and the statutory-minimum automatic-enrolment workplace pension. Here too a boundary convention is required: the employee and employer contribution indicators for the United Kingdom combine National Insurance (employee at 8 percent then 2 percent between the thresholds; employer at 15 percent above 5,000 pounds per year, uncapped) with the auto-enrolment pension at the statutory minimum rate (5 percent employee and 3 percent employer of qualifying earnings between 6,240 and 50,270 pounds per year). This is a broader perimeter than social contributions alone, and it departs from the treatment applied to the other countries; the choice is made because auto-enrolment is a legal obligation, and it is flagged so that the resulting wedge is not read as a pure social contribution wedge.

Income tax is computed separately via the progressive rUK schedule: 0 percent up to the personal allowance of 12,570 pounds, tapered by 1 pound for every 2 pounds of income above 100,000 pounds until extinction at 125,140 pounds, then 20, 40 and 45 percent. Only the bands applicable to England, Wales and Northern Ireland are

modelled; the separate Scottish bands are not.

6. The Comparison Module

The comparison module places the six national modules on a common wage grid. It redefines no national parameter and introduces no new institutional assumption: it is a harmonization and orchestration layer.

6.1. Harmonized wage grid

The harmonized grid contains 21 wage points, from 2,000 to 12,000 international dollars per month, with a step of 500. The lower bound is deliberately set below the minimum or reference wage floor of all six countries once converted into international dollars, so that the lowest wage ranges remain visible rather than being truncated. Wage points falling below a country's statutory minimum are flagged as hypothetical for that country rather than silently dropped, which preserves the continuity of the curves while making the flagged region explicit.

Table 2: Statutory minimum or reference wage thresholds, converted to international dollars (approximate)

Country	Threshold (intl \$/month)
France (SMIC)	2,604
Switzerland	2,710
Belgium (RMMM)	2,911
United Kingdom (National Living Wage)	3,025
Netherlands (WML)	3,152
Germany (Mindestlohn)	3,434

6.2. Purchasing power parity conversion

Each harmonized wage point $\tilde{w}$ is converted into the local currency of country c using a purchasing power parity conversion factor π_c :

$$w_c = \pi_c \cdot \tilde{w}. \quad (7)$$

The factor retained is the household private-consumption purchasing power parity conversion factor published by the World Bank under the International Comparison Program, indicator PA.NUS.PRVT.PP, 2024 reference year (15). It is applied indi-

vidually to all six countries, including Switzerland, following the same logic as the OECD's *Taxing Wages* comparisons (10).

The choice of a consumption-based rather than a GDP-based parity is deliberate: the object of comparison is the standard of living attainable from a wage, not aggregate output per capita.

6.3. Direct calculation without interpolation

For each local wage w_c obtained from equation (7), the calculation function of the corresponding national module is called directly. Values are not obtained by interpolating already-computed national datasets. This matters because the schedules being compared are precisely the ones containing kinks: interpolation across a relief phase-out threshold or a tax bracket boundary would smooth away the feature the comparison is meant to expose.

6.4. Reference profiles

Each country is evaluated at a single documented reference profile.

Table 3: Reference profiles by country

Country	Reference profile
France	Non-executive, outside Alsace-Moselle, firm with fewer than 50 employees, standard AT/MP rate, full-time
Germany	Public health insurance, childless, outside Saxony
Belgium	Private-sector employee (only profile currently modelled)
Switzerland	Population-weighted average across the 26 cantons, A0N tariff (single, no children, no church tax), standardized LPP and accident-insurance assumptions
Netherlands	Permanent contract (low WW rate), large employer (high Aof rate), single, no children, below AOW age
United Kingdom	Standard employee outside Scotland (rUK bands), full-time, standard tax code, auto-enrolled at the statutory minimum rate

6.5. Income tax treatment in comparison

Including France in the after-tax indicators requires a convention, because the French income tax is assessed annually at household level rather than withheld at a schedule-determined rate on the individual wage. The module applies the actual progressive

schedule to a single, childless taxpayer with one tax unit: the monthly net taxable salary computed by the Mon-entreprise engine is annualized, reduced by the standard 10 percent professional-expense allowance within its floor and ceiling, run through the marginal brackets, and converted back to a monthly amount.

This is directly comparable to the treatment of the other five countries. German Lohnsteuer, Belgian precompte professionnel, Swiss withholding tax, Dutch loonheffing and British PAYE are all mandatory levies computed from the wage for a single, childless profile. The French progressive schedule computes, in the same way, the tax actually due for that same profile type.

The convention is nonetheless restrictive and is flagged as such: it assumes no other income, no deductible charge and no tax credit or reduction, and it uses only the scales applicable to metropolitan France.

6.6. Comparative indicators

The comparison module reports, at each harmonized wage point and for each country: net wage before tax, net wage after tax, employer cost, employee contributions, employer contributions, the social wedge before tax as a rate, the total wedge after tax as a rate, the employer cost to net-before-tax ratio and the employer cost to net-after-tax ratio, in both local currency and international dollars. The dataset is downloadable, together with the parameter and source files for the comparison year.

7. Indicators

This section defines the indicators produced by the Lab. All indicators are defined at the level of a given wage point and profile.

7.1. Level indicators

Gross monthly wage (w_{brut}) is the contractual gross wage before any employee social contribution is deducted.

Net monthly wage before tax is defined as:

$$w_{\text{net}} = w_{\text{brut}} - \text{Cot}_{\text{sal}}. \quad (8)$$

Total employer cost is defined as:

$$C_{\text{emp}} = w_{\text{brut}} + \text{Cot}_{\text{emp}}^{\text{net}}. \quad (9)$$

Employee contributions (Cot_{sal}) are all social contributions legally assigned to the employee and deducted from gross wage, subject to the national perimeter conventions of Sections 5.4 and 5.5.

Employer contributions net of relief ($\text{Cot}_{\text{emp}}^{\text{net}}$) are all contributions legally assigned to the employer after deduction of the applicable statutory reduction.

RGDU 2026 amount is, for France, the monthly employer contribution relief, also expressed as a share of gross wage.

7.2. Rate indicators

The effective employer contribution rate is:

$$\tau_{\text{emp}}^{\text{eff}} = \frac{\text{Cot}_{\text{emp}}^{\text{net}}}{w_{\text{brut}}}. \quad (10)$$

This is an apparent rate after reliefs. It is substantially lower than the statutory rate at wages near the SMIC and converges toward the statutory rate as the RGDU phases out.

The total social levy rate is:

$$\tau_{\text{total}}^{\text{brut}} = \frac{\text{Cot}_{\text{emp}}^{\text{net}} + \text{Cot}_{\text{sal}}}{w_{\text{brut}}}. \quad (11)$$

The same quantity is also reported relative to total employer cost, which is the convention used in the comparison module:

$$\tau_{\text{total}}^{\text{cost}} = \frac{\text{Cot}_{\text{emp}}^{\text{net}} + \text{Cot}_{\text{sal}}}{C_{\text{emp}}}. \quad (12)$$

7.3. Wedge and cost-to-net ratio

The socio-fiscal wedge before tax is $W = C_{\text{emp}} - w_{\text{net}}$, as in equation (5), and the total wedge after tax is W^{tot} , as in equation (6).

The employer cost-to-net-wage ratio is:

$$\rho = \frac{C_{\text{emp}}}{w_{\text{net}}}. \quad (13)$$

This indicator answers a practical question: how many euros – or international

dollars – does the employer pay for one unit of net wage received by the employee?

7.4. Marginal employer cost

Marginal indicators are computed by finite difference between adjacent wage points i and $i + 1$. Let Δ denote this discrete difference operator. The marginal employer cost rate is:

$$\tau_{\text{emp}}^{\text{marg}} = \frac{\Delta C_{\text{emp}}}{\Delta w_{\text{brut}}}. \quad (14)$$

In the RGDU phase-out zone, this rate can exceed the average effective contribution rate, because a wage increase both raises statutory contributions and reduces the relief received.

7.5. Socio-fiscal return on a gross wage increase

The baseline pre-tax and pre-benefit return is:

$$R^{\text{net}} = \frac{\Delta w_{\text{net}}}{\Delta w_{\text{brut}}}. \quad (15)$$

The reference socio-fiscal disposable income is:

$$Y^{\text{sf}} = 12w_{\text{net}} - \text{IR}^{\text{est}} + \text{PA}^{\text{est}}, \quad (16)$$

where IR^{est} is estimated annual personal income tax and PA^{est} is the estimated annual prime d'activite for the reference profile. The socio-fiscal return is then:

$$R^{\text{sf}} = \frac{\Delta Y^{\text{sf}}/12}{\Delta w_{\text{brut}}}. \quad (17)$$

The difference between R^{net} and R^{sf} captures the marginal effect of the tax-benefit system on top of employee social contributions:

$$E^{\text{sf}} = R^{\text{net}} - R^{\text{sf}}. \quad (18)$$

A positive value means that income tax and benefit withdrawal reduce the return to a gross wage increase. A negative value may arise locally if the benefit formula or a threshold effect increases disposable income more than the net wage alone would suggest. Because the calculation is based on finite differences over a fine wage grid,

visible kinks should be interpreted as institutional threshold effects rather than as smooth behavioural responses.

7.6. The FLCL Index

The France module reports a family of labour cost efficiency indicators, retained under the FLCL name. The core index measures the share of employer cost that reaches the employee as net wage:

$$\text{FLCL-E} = 100 \times \frac{w_{\text{net}}}{C_{\text{emp}}}. \quad (19)$$

The index is complemented by a gross-wage counterpart, FLCL-B, which applies the same efficiency logic to the gross wage rather than the net wage, and by FLCL-R, which isolates the contribution of general reliefs to wage efficiency by comparing the index with its counterfactual value computed without the relief.

Three marginal indicators complete the family:

- **Marginal transmission:** the share of one additional euro of employer cost that reaches the net wage;
- **Marginal capture:** the complementary share absorbed by employee and employer contributions;
- **Implicit progressivity:** the variation of FLCL-E as the wage level increases, which summarizes whether the system becomes more or less efficient at transmitting cost into net wage along the distribution.

A low-wage support indicator reports the magnitude of the relief-driven gap between the observed index and its no-relief counterfactual in the lower part of the wage scale. Together, these indicators express in a single scale-free metric what the level and rate indicators express separately, which makes them convenient for cross-profile and, prospectively, cross-country comparison.

7.7. Employer cost decomposition

The employer cost decomposition breaks total employer cost into three components:

$$C_{\text{emp}} = \underbrace{w_{\text{net}}}_{\text{net wage}} + \underbrace{\text{Cot}_{\text{sal}}}_{\text{employee contributions}} + \underbrace{\text{Cot}_{\text{emp}}^{\text{net}}}_{\text{employer contributions after relief}}. \quad (20)$$

The relief amount is displayed explicitly by showing the gap between theoretical employer contributions before relief and effective employer contributions after relief:

$$\text{Cot}_{\text{emp}}^{\text{gross}} = \text{Cot}_{\text{emp}}^{\text{net}} + \text{RGDU}. \quad (21)$$

This representation makes the mechanical role of contribution reliefs immediately visible.

8. Descriptive Results

This section provides a descriptive commentary of the main outputs. All patterns described below are descriptive and institutional. They should not be interpreted as causal estimates.

8.1. From gross wage to employer cost

The gap between net wage and employer cost is not proportional to gross wage. At wages near the SMIC, contribution reliefs compress this gap: employer cost is only moderately above gross wage because the RGDU reduces net employer contributions. As wages increase and the RGDU phases out, the gap between employer cost and net wage widens progressively.

For the baseline French profile – non-executive, outside Alsace-Moselle, firm with fewer than 50 employees, standard AT/MP rate, full-time – the module reports a net wage of about 1,429 euros and a total employer cost of about 1,992 euros at one SMIC, with an RGDU of about 700 euros per month, and an employer cost to net wage ratio of about 1.80 at two SMIC. These values are conditional on the parameter version declared in the module and should be re-read whenever that version changes.



Figure 1: From gross wage to total employer cost, France module

8.2. Effective employer contribution rate

The effective employer contribution rate displays a non-linear profile. At the SMIC level, the rate is substantially below the statutory rate due to the full RGDU relief. As wages increase above the SMIC, the effective rate rises. Above the relief extinction zone, it stabilizes close to the statutory employer contribution schedule. The extension of the grid to 6.0 SMIC makes this convergence directly observable rather than merely asserted.

8.3. RGDU 2026 and the 1 June 2026 revalorisation

The RGDU graph plots the monthly relief amount and its share of gross wage. The relief is maximal at one SMIC and decreases thereafter, reaching zero in the upper part of the degressive zone.

The 1 June 2026 minimum wage revalorisation provides an instructive natural configuration, which the module treats as an explicit before-and-after comparison. A degressive relief formula is anchored on the SMIC. When the SMIC is revalorised while the relief parameters are held constant, the entire schedule is displaced: at a given gross wage, the wage-to-SMIC ratio falls, but the relief entitlement does not mechanically follow, so the relief extinguishes earlier in absolute wage terms and employer cost rises across a range of the distribution. The module reports a marked increase in employer cost concentrated around three SMIC, corresponding to the

earlier phase-out of the relief under frozen parameters.

This exercise is exploratory and rests on currently available Mon-entreprise/URSSAF outputs. It is nonetheless the clearest available illustration of a general point: the interaction between an indexed wage floor and a non-indexed relief schedule is itself a policy parameter, and one whose incidence is concentrated well above the minimum wage.



Figure 2: RGDU 2026 relief along the wage scale

8.4. Socio-fiscal wedge

The wedge increases in absolute terms with gross wage. As a share of employer cost or gross wage, it is compressed at low wages by the RGDU and rises as the relief phases out. This non-constant relative wedge is one of the key features of the French contribution architecture, and it is the feature that a single headline rate is least able to convey.

8.5. Marginal employer cost

The marginal employer cost indicator reveals that the phase-out of the RGDU raises the local cost of wage increases. In the degressive zone, employers face a marginal

cost of gross wage increases that exceeds the average effective rate and diverges from what a simple statutory schedule would suggest.

8.6. Socio-fiscal return on a gross wage increase

The socio-fiscal return graph shows how much of an additional euro of gross wage is converted into disposable income for the reference profile. The pre-tax and pre-benefit return reflects employee social contributions; the socio-fiscal return adds estimated personal income tax and estimated prime d'activite.

This graph is especially informative around low and middle wages. Near the SMIC, the prime d'activite significantly alters the marginal return to work. As earnings increase, the benefit is progressively withdrawn, which reduces the marginal disposable return of wage increases. At higher levels of income, the income tax schedule becomes the dominant factor. The graph therefore visualizes the interaction between payroll contributions, income taxation and benefit withdrawal along the earnings scale.

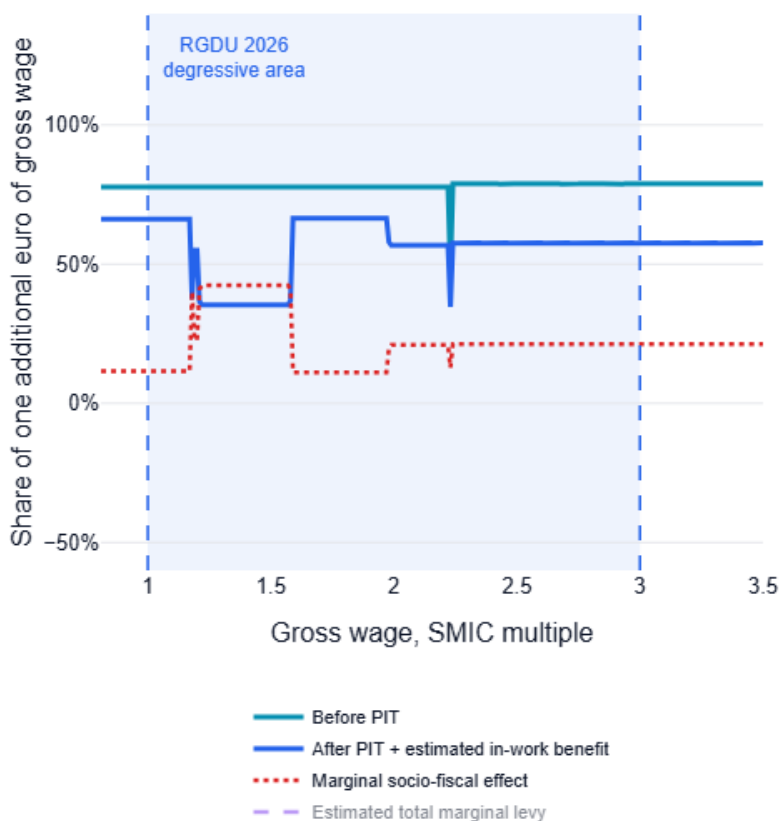


Figure 3: Socio-fiscal return of a gross wage increase

8.7. Employer cost decomposition

The decomposition graph displays the components of employer cost at selected wage points: net wage, employee contributions, employer contributions after relief and the relief amount. This graph is useful pedagogically because it makes the distinction between gross wage, net wage and employer cost immediately visible.

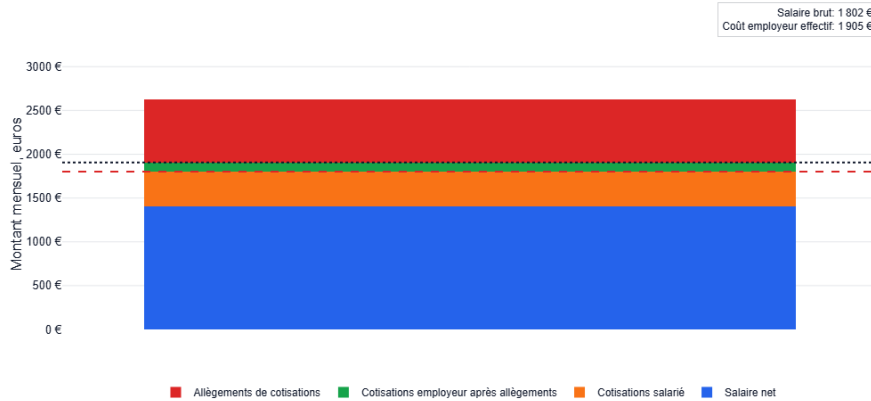


Figure 4: Employer cost decomposition at one SMIC

8.8. Within-country comparisons

Within the France module, the most instructive comparisons are:

- **Executive versus non-executive employees:** the two profiles follow broadly similar patterns but diverge at mid-range wages due to differences in contribution structures, notably around the social security ceiling.
- **AT/MP scenarios:** varying the occupational injury risk rate shifts the employer cost schedule while preserving the general shape of the relief profile.
- **Territorial regime:** the Alsace-Moselle regime modifies both employee-side and employer-side contribution structures, producing a distinct wedge decomposition at equal gross wage.
- **Firm size and working time:** these dimensions shift the level of employer cost and, in the part-time case, relocate the profile within the relief schedule.

8.9. Cross-country patterns

The comparison module reports, for each country and each harmonized wage point, the share of employer cost reaching the employee as net wage. Read at the level of the whole grid rather than at a single point, three patterns emerge.

First, the ordering of countries by pre-tax wedge is not the ordering by post-tax

wedge, because systems differ in the split between social contributions and income taxation. Any statement about the relative burden on labour is therefore conditional on which wedge is being reported, and the Lab reports both.

Second, the countries with the highest employer-side contribution levels are also those operating low-wage employer reduction schemes, so their effective schedules are more strongly non-linear along the wage scale than their statutory rates suggest.

Third, the perimeter conventions documented in Sections 5.4 and 5.5 are quantitatively material: the Dutch zero for employee contributions and the British inclusion of auto-enrolment pension both move the reported wedge substantially, and neither is an economic result. This is the central methodological warning attached to the comparison module.

9. Dashboard Architecture

The Lab is deployed as a set of static interactive pages, accessible at:

<https://hugosragain.github.io/labor-cost-lab/>

Landing page. Presents the project, a schematic decomposition from employer cost to net wage, and an entry point to each national module together with its maturity status.

National modules. Each module page follows the same tab structure: simulation, comparisons, data, methodology and contact. The France module adds a dedicated FLCL Index tab.

Comparison module. Organized in three tabs – comparison at a given wage point, comparative charts across the whole harmonized grid, and methodology – with a data section exposing the comparison dataset, the parameter file and the sources file.

Publications. The site hosts the working papers of the project: the France module methodology, the framework document for the international extension, and the present paper.

Data and reproducibility. A dedicated section exposes the four artefact families: computed wage grids in CSV, versioned regulatory parameters in JSON, documented sources, and open code.

10. Intended Uses

Applied economics research. The Lab provides reproducible datasets of labour cost indicators at fine wage resolution, suitable for descriptive studies of contribution reliefs, effective rate schedules and marginal cost profiles, in one country or across several.

Policy analysis. Analysts can identify wage ranges where marginal costs, effective rates or socio-fiscal marginal effects are particularly high or low, and can assess the displacement effects of a change in an indexation parameter.

Comparative institutional analysis. The comparison module supports structured claims about how different systems allocate the same employer cost between net wage, contributions and taxation at equivalent standards of living.

Pedagogy. The decomposition graph, the FLCL Index and the socio-fiscal return graph make complex institutional mechanisms legible in courses on labour economics, social policy and public finance.

Threshold and non-linearity analysis. The wage grid enables identification of kinks generated by reliefs, tax brackets and benefit withdrawal.

11. Methodological Limitations

Reference simulations, not official payroll calculations. Actual payroll values may differ depending on collective agreements, firm size, sector, precise risk classification, exemptions and specific contribution regimes not covered by the general simulation framework.

Heterogeneous module maturity. France is the reference implementation and Belgium a full 2026 module; Germany, Switzerland, the Netherlands and the United Kingdom are documented prototypes. Comparative statements should be weighted accordingly.

Perimeter conventions are not neutral. The Dutch bundling of national insurance into the income tax schedule and the British inclusion of statutory auto-enrolment pension both alter the measured wedge for reasons that are conventional rather than economic. They are documented and flagged, not corrected away.

Dependence on external engines and parameters. If the Mon-entreprise/URSSAF engine or any national parameter set is updated, the corresponding dataset must be regenerated. Users relying on the data for time-sensitive analysis should verify the regulatory reference period and the declared parameter version.

Purchasing power parity is an approximation. Consumption-based parity factors are estimated, revised, and averaged over national consumption baskets. They reduce, but do not eliminate, the incomparability of nominal wages, and they are not adjusted for within-country regional price dispersion.

Single reference profile per country. Effective rates depend on household configuration. The reference profiles are explicit conventions – broadly single, childless, full-time – and do not represent population averages.

No behavioural responses or economic incidence. The Lab models accounting and institutional schedules only. It does not estimate employment effects, wage bargaining responses, labour supply responses, or the economic incidence of contributions and benefits.

No complete socio-fiscal microsimulation. The socio-fiscal module is deliberately restricted to a reference profile. It does not model the diversity of household configurations, housing benefits, family benefits, local benefits, wealth effects, tax credits, tax reductions or real-expense deductions. Only metropolitan France scales are used.

Static framework. The Lab provides a cross-sectional picture at a given regulatory state. It does not simulate dynamic effects of parameter changes, although the pipeline can be regenerated when parameters evolve, and the June 2026 before-and-after exercise illustrates how such regeneration can be exploited analytically.

12. Conclusion

The *Labor Cost Lab* transforms a set of complex institutional systems into transparent and reproducible objects of economic analysis. Starting from the *French Labour Cost Lab*, whose architecture separated a wage grid, a parameter set and an indicator layer, the project generalizes that separation into a multi-country platform: six national modules producing harmonized outputs, and a comparison module that places them on a wage grid expressed in international dollars at purchasing power parity.

The Lab makes four categories of result visible. First, it shows how total employer cost, net wage and the socio-fiscal wedge diverge along the wage scale in ways not apparent from aggregate indicators or isolated case-type calculations. Second, it reveals the non-linear structure of effective employer contribution rates introduced by degressive relief mechanisms, and quantifies how a change in the indexation of the wage floor displaces the entire relief schedule. Third, it documents how income taxation and in-work benefits modify the marginal disposable return to gross wage increases. Fourth, it exposes which cross-country differences are institutional and which are artefacts of perimeter conventions, currency conversion or price levels.

By design, the Lab does not claim to estimate the economic incidence of contributions or the employment effects of relief policies. It provides the institutional and computational infrastructure on which such analyses can subsequently be built. The *Labor Cost Lab* is open-source, versioned and designed for reproducibility. It is intended as a living tool, to be updated as regulatory parameters evolve and as further national modules reach maturity.

A. Notation Summary

Table 4: Summary of notation

Symbol	Definition
w_{brut}	Gross monthly wage
w_{net}	Net monthly wage before income tax
C_{emp}	Total employer cost
Cot_{sal}	Employee social contributions
$\text{Cot}_{\text{emp}}^{\text{gross}}$	Employer contributions before relief
$\text{Cot}_{\text{emp}}^{\text{net}}$	Employer contributions after relief
RGDU	Reduction Generale Degressive Unique relief (France)
IR^{est}	Estimated annual personal income tax
PA^{est}	Estimated annual prime d'activite
Y^{sf}	Estimated annual socio-fiscal disposable income
W	Socio-fiscal wedge before tax
W^{tot}	Total wedge after tax
$\tau_{\text{emp}}^{\text{eff}}$	Effective employer contribution rate
$\tau_{\text{emp}}^{\text{marg}}$	Marginal employer cost rate
$\tau_{\text{total}}^{\text{brut}}$	Total levy rate as a share of gross wage
$\tau_{\text{total}}^{\text{cost}}$	Total levy rate as a share of employer cost
ρ	Employer cost-to-net-wage ratio
R^{net}	Net wage return on a gross wage increase
R^{sf}	Socio-fiscal return on a gross wage increase
π_c	Purchasing power parity conversion factor for country c
$\tilde{w}$	Harmonized wage point in international dollars

B. France Module: Profile Configuration

Table 5: Profile dimensions, France module

Dimension	Modalities	Count
Employee status	Non-executive / Executive	2
Territorial regime	Standard / Alsace-Moselle	2
AT/MP risk scenario	Standard / 1% / 4% / Support functions	4
Core profiles (downloadable dataset)		16
Firm size (interface parameter)	< 50 / $\geq$ 50 employees	2
Working time (interface parameter)	Full-time / 80% / 50%	3
Total interface configurations		96

C. Wage Grid Parameters

Table 6: Wage grid configuration

Parameter	Value
<i>France module</i>	
Minimum point	0.8 SMIC
Maximum point	6.0 SMIC
Step	0.01 SMIC
Number of wage points	521
Core profiles	16
Observations in core dataset	8,336
Regulatory parameter version	2026-06
Effective date	1 June 2026
<i>Comparison module</i>	
Minimum point	2,000 intl \$
Maximum point	12,000 intl \$
Step	500 intl \$
Number of wage points	21
Countries	6
Observations	126
PPP reference year	2024

D. Reference SMIC Values, 2026

Table 7: Gross monthly SMIC, 35-hour full-time equivalent

Effective date	Hourly gross	Monthly gross
1 November 2024	11.88 euros	1,801.80 euros
1 January 2026	12.02 euros	1,823.03 euros
1 June 2026	12.31 euros	1,867.02 euros

The 1 June 2026 revalorisation of 2.41 percent results from the automatic mechanism of article L. 3231-5 of the Labour Code, triggered when the consumer price index for the lowest-income households rises by at least 2 percent since the last determination of the minimum wage, and is fixed by the arrete of 22 May 2026 ([Arrete du 22 mai 2026](#)).

E. Reference Socio-Fiscal Module (France)

The reference socio-fiscal module is computed for a single individual with one tax unit, no children, no other income and no housing benefit.

Let w_{net} be the net monthly wage. Annual net wage is:

$$w_{\text{net}}^{\text{ann}} = 12 \cdot w_{\text{net}}.$$

Taxable employment income is approximated by applying the standard 10 percent professional-expense allowance, bounded by its statutory floor and ceiling:

$$R_{\text{taxable}} = w_{\text{net}}^{\text{ann}} - \min[\max(0.10 \cdot w_{\text{net}}^{\text{ann}}, 509), 14,555].$$

Estimated personal income tax is obtained by applying the 2026 progressive income tax schedule to R_{taxable} for one tax unit, including the decote and the low-tax recovery threshold:

$$\text{IR}^{\text{est}} = f_{\text{IR}}(R_{\text{taxable}}).$$

The estimated monthly prime d'activite is approximated as:

$$\text{PA}_{\text{month}}^{\text{est}} = \max[0, F + \alpha w_{\text{net}} + B(w_{\text{net}}) - w_{\text{net}}],$$

where F is the monthly forfait for a single person, α the earned-income coefficient and $B(w_{\text{net}})$ the individual bonus as a function of monthly earned income. In the reference scenario, no housing forfait, no other resources and no family benefits are included.

Annual socio-fiscal disposable income is then:

$$Y^{\text{sf}} = w_{\text{net}}^{\text{ann}} - \text{IR}^{\text{est}} + 12 \cdot \text{PA}_{\text{month}}^{\text{est}}.$$

This module is indicative. It is designed to illustrate marginal interactions between income taxation and in-work benefits, not to replace the official tax administration or CAF simulators.

F. Comparison Module: Country Conventions

Table 8: Income tax and contribution perimeter by country

Country	Convention
France	Progressive schedule (0/11/30/41/45 percent), 2025 income taxed in 2026, 10 percent allowance with floor and ceiling, one tax unit
Germany	Lohnsteuer; public health insurance, childless, outside Saxony
Belgium	Precompte professionnel; ONSS contributions and employer structural reduction
Switzerland	Cantonal withholding tax, tariff A0N, population-weighted average across 26 cantons
Netherlands	Loonheffing, Box 1 2026 schedule with algemene heffingskorting and arbeidskorting; employee national insurance bundled into the first bracket, hence employee contributions reported as zero
United Kingdom	PAYE (rUK bands only); contribution indicators combine Class 1 National Insurance and statutory-minimum auto-enrolment pension

References

- [Arrete du 22 mai 2026] Arrete du 22 mai 2026 relatif au relèvement du salaire minimum de croissance. *Journal officiel de la Republique francaise*, 24 May 2026.
- [2] Bozio, A., and Wasmer, E. (2024). *Les politiques d'exonerations de cotisations sociales: une inflexion necessaire*. Rapport remis au Premier ministre, Paris.
- [3] Cahuc, P., Carcillo, S., and Zylberberg, A. (2019). *Labor Economics*, 2nd edition. MIT Press, Cambridge, MA.
- [4] Federal Statistical Office. (2024). *Permanent resident population by canton*. Neuchatel. <https://www.bfs.admin.ch>
- [5] Gruber, J. (1997). The incidence of payroll taxation: evidence from Chile. *Journal of Labor Economics*, 15(3), S72–S101.
- [6] Kotlikoff, L. J., and Summers, L. H. (1987). Tax incidence. In A. J. Auerbach and M. Feldstein (eds.), *Handbook of Public Economics*, Vol. 2, pp. 1043–1092. Elsevier, Amsterdam.
- [7] Kramarz, F., and Philippon, T. (2001). The impact of differential payroll tax subsidies on minimum wage employment. *Journal of Public Economics*, 82(1), 115–146.
- [8] Kugler, A., and Kugler, M. (2009). Labor market effects of payroll taxes in developing countries: evidence from Colombia. *Economic Development and Cultural Change*, 57(2), 335–358.
- [9] L'Horty, Y., and Roux, S. (2003). Vingt ans d'allegements de cotisations sociales en France: une evaluation. *Revue de l'OFCE*, 87, 19–60.
- [10] OECD. (2026). *Taxing Wages*. OECD Publishing, Paris.
- [11] Saez, E., Matsaganis, M., and Tsakloglou, P. (2012). Earnings determination and taxes: evidence from a cohort-based payroll tax reform in Greece. *Quarterly Journal of Economics*, 127(1), 493–533.
- [12] Spring-Ragain, H. (2026a). *French Labour Cost Lab: An Open-Source Dashboard for Simulating Labour Costs, Contribution Reliefs and Socio-Fiscal Marginal Returns in France*. Working Paper, CEDS Paris.
- [13] Spring-Ragain, H. (2026b). *Labor Cost Lab*. Open-source comparative observatory, CEDS Paris. <https://hugosragain.github.io/labor-cost-lab/>

- [14] URSSAF. (2026). *Mon-entreprise: public calculation engine and simulator*.
<https://mon-entreprise.urssaf.fr>
- [15] World Bank. (2024). *International Comparison Program: PPP conversion factor, private consumption* (indicator PA.NUS.PRVT.PP). Washington, DC.
<https://data.worldbank.org>
- [16] Caisse nationale des allocations familiales. (2026). *Prime d'activite: baremes et conditions d'eligibilite*. <https://www.caf.fr>
- [17] Service Public. (2026). *Impot sur le revenu: bareme et calcul*. <https://www.service-public.fr>